

Problem

A certain band-limited periodic current has only three frequencies in its Fourier series representation: dc, 50 Hz, and 100 Hz. The current may be represented as

$$i(t) = 4 + 6 \sin 100\pi t + 8 \cos 100\pi t \\ - 3 \sin 200\pi t - 4 \cos 200\pi t \text{ A}$$

(a) Express $i(t)$ in amplitude-phase form.

(b) If $i(t)$ flows through a $2\text{-}\Omega$ resistor, how many watts of average power will be dissipated?

Step-by-step solution

Step 1 of 2

$$(A) \quad A_n = \sqrt{a_n^2 + b_n^2}, \quad \phi = \tan^{-1} \left(\frac{b_n}{a_n} \right)$$

$$A_1 = \sqrt{6^2 + 8^2} = 10, \quad \phi_1 = \tan^{-1} \left(\frac{6}{8} \right) \\ = 36.87^\circ$$

$$A_2 = \sqrt{3^2 + 4^2} = 5, \quad \phi_2 = \tan^{-1} \left(\frac{3}{4} \right) \\ = 36.87^\circ$$

$$i(t) = \left\{ 4 + 10 \cos(100\pi t - 36.87^\circ) - 5 \cos(200\pi t - 36.87^\circ) \right\} \text{ A}$$

Step 2 of 2

$$(B) \quad p = I_{dc}^2 R + 0.5 \sum I_n^2 R$$

$$= 2 \left[4^2 + 0.5(10^2 + 5^2) \right] \\ = 157 \text{ W}$$